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Land use affects dung beetle communities and their ecosystem service in forests and grasslands

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A B S T R A C T

Dung beetles (Scarabaeidae) are common detritivores, and especially the tunnelling genera *Geotrupes*, *Anoplotrupes* and *Onthophagus* enhance the soil quality and support nutrient cycles by rapid burial of mammalian dung. These functionally important beetles are faced with a wide range of anthropogenic disturbances and changes in environmental conditions due to land use. We thus conducted quantitative surveys of the abundance (converted to total biomass) of dung beetles and their dung removal rates (g per two days) in 150 forest and 150 grassland sites with varying land-use intensity, located in north-east, central and south-west Germany. We used dung from livestock (cow, sheep, horse) and game animals (wild boar, red deer and fox) to provide a characteristic spectrum of dung resources on each site. Most dung beetle species showed habitat preferences: *Anoplotrupes*, *Typhaeus* and several *Aphodius* species almost exclusively occurred in forests, while most *Onthophagus* individuals were found in grasslands. In total we collected 18780 individuals from 33 species. The average dung beetle biomass was 36 times higher in forests than in grasslands, and their effective dung removal rate was 3 times increased. The beetles' total biomass was strongly correlated to their removal rates. In forests, the amount of wood harvesting significantly reduced dung removal rates by 20%, and mowing frequency (−7%) and fertilisation (−4%) had a significant negative effect in grasslands. Dung removal by beetles increased with grazing intensity (+6%), however, and was higher in non-native coniferous forests (+22%). Overall, our study demonstrates negative effects of habitat conversion from forest to grassland, and negative effects of land-use intensity within forests and grasslands on dung beetle activities.

1. Introduction

Fossil evidence suggests that dung beetles exist since the Mesozoic Era (Late Jurassic − Early Cretaceous) thus demonstrating that the usage of dung became an efficient strategy of resource acquisition in a very early stage of fauna evolution (Chin and Gill, 1996; Davis et al., 2002; Nikolajev and Dong, 2010). Furthermore the nearly cosmopolitan superfamily of Scarabaeoidea are the only known invertebrates that store fecal material in tunnels (Vander Wall, 1990).

Despite choosing an unpredictable and patchy occurring resource, dung consumption grants sufficient nutrients for adults and beetle larvae (Philips, 2011). Because of their tunnelling behavior, dung beetles increase the input of nutrients into the soil, benefit the vegetation (Nichols et al., 2008; Wu et al., 2011) and minimize potential breeding grounds of (pathogenic) pests (Fincher, 1973; Ridsdill-Smith and Edwards, 2011). Dung pads would remain much longer without dung beetle activity (Walters, 2008), preventing growth of vegetation and therefore may result in wasted pastures up to two years (Anderson et al., 1984). Additionally burial of dung causes soil aeration and access for water (Bornemissza, 1960; Bang et al., 2005), and it decreases soil compaction (Manning et al., 2016).

Although dung beetles are considered as generalists regarding their resources, various reactions towards their preference for dung types have been shown (Hanski and Cambefort, 1991). Whether it depends on the "host animals" diet (carnivore, herbivore, omnivore) (Halffter and Matthews, 1966; Whipple and Hoback, 2012), nutrients (Whipple and Hoback, 2012), odour intensity (Scholtz et al., 2009) or differences in volatile organic compounds (Schmitt et al., 2004; Dormont et al., 2007) − dung beetles are attracted to a wide range of different dung types, but in variable numbers (Whipple and Hoback, 2012).

In spite of their ubiquitous presence, several dung beetle species are habitat-specific, and forests and grassland communities differ substantially (Roslin and Viljanen, 2011). The beetles' sensitivity to disturbances varies across species, rendering dung beetles as suitable

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biological indicators often considered in monitoring programs, supported by the fact that their sampling is very simple and efficient (Scholtz et al., 2009). Several authors surveyed the diversity of dung beetles in response to land use in tropical ecosystems, where they show their highest diversity in forests and savannas (Hanski and Cambefort, 1991; Estrada and Coates-Estrada, 2002; Feer and Hingrat, 2005; Hanski et al., 2007; Nichols et al., 2007; Barragan et al., 2011). In Europe, anthropogenic influences on the diversity and occurrence of dung beetles have also been monitored (Martín-Piera and Lobo, 1995; Hutton and Giller, 2003; Spector, 2006; Zamora et al., 2007), highlighting changes for certain regions and habitats, such as grasslands and shrubs versus planted forests (Romero-Alcaraz and Avila, 2000; Tocco et al., 2013). However, management activities are particularly diverse in European cultural landscapes, including different silvicultural management types, farm types (conventional versus organic), production systems (cropland, grassland, fertilisation and livestock) and socio-economic conditions (Reidsma et al., 2006). Apart from a differentiation in the type of land use, their quantitative intensities are strongly variable (Herzog et al., 2006). Reviews at the global scale (Newbold et al., 2015) thus highlighted the importance of local-scale studies for suitable characterisation and projection of land use on (local) biodiversity and ecosystem services (Allan et al., 2015; Soliveres et al., 2016). In our study, we thus focused on such local effects of continuous land-use intensity gradients in forests and in grasslands on dung beetle abundance and ecosystem functioning. In addition to land-use gradients within forests or grasslands, we explicitly compared forest versus grassland, representing the two most common habitat types apart from arable fields and reflecting the historical habitat conversion from unmanaged or managed forest, originally covering vast parts of Central Europe, to cultivated grassland. In particular, we assessed (a) the abundance (biomass) of dung beetles, which are potentially involved in removal of various types of dung, and (b) the dung removal rate by these beetles within their habitats. Our goal was to quantify (1) how forests and grasslands differ in dung beetle biomass and their removal activities, (2) how habitat-specific, gradual variation in land-use intensity affects these beetles and their removal rates and (3) to understand which components of land use are responsible for this variation.

2. Material and methods

2.1. Study site

We conducted our study within the framework of the Biodiversity Exploratories project, comprising a large number of representative forest and grassland sites in three regions (north-east, central and south-west Germany) (Fischer et al., 2010). These sites varied continuously in land-use intensity, which was quantified based on farmer interviews and forest surveys. The three regions are: (1) Biosphere Reserve Schorfheide-Chorin (SCH; in North-East Germany, ~13.000 km^2, 3–140 m a.s.l., 13°23′27″–14°08′53″ E/52°47′25″–53°13′26″ N), (2) Hainich National Park and surroundings (HAI; in Central Germany, ~13.000 km^2, 285–550 m a.s.l., 10°10′24″–10°46′45″ E/ 50°56′14″–51°22′43″ N) and (3) Biosphere Reserve Schwäbische Alb (ALB; in South-West Germany, ~422 km^2, 460–860 m a.s.l., 09°10′49″–09°35′54″ E/48°20′28″– 48°32′02″N). Using a grid of 100 × 100 m placed over the entire area within each region, experimental plots (hereafter: sites) were chosen at random. Sites with inhomogeneous land cover or partial overlap with settlements, agricultural fields, water bodies and sites intersected by roads were discarded. In each region, 100 square-shaped sites were selected, 50 sites in forests (each 100 × 100 m) and 50 in grasslands (50 × 50 m), which are representative for the regional variation in land-use and management intensities. All sites are surrounded by a larger area of the same land use, i.e. the squares are usually only a small part of the forest or grassland with a specific management.

Our studies are based on two approaches:

- Comprehensive survey: a survey of all 300 experimental sites during summer 2014 was conducted once to maximize spatial replication.
- Intensive survey: on a subset of 54 of these sites (9 forests and 9 grasslands per region), we repeatedly surveyed the dung beetles and their activity to account for temporal variation across seasons and years. Since the comprehensive survey includes these 54 sites, we additionally used this subset from summer 2014 (a) in the analyses of temporal variation.

For the comprehensive survey we sampled the 100 sites per region in 20 days (SCH – June, HAI – July, ALB – August) (10.06.14–04.07.14; 07.07.14–01.08.14; 04.08.14–29.08.14). For the intensive survey we sampled each region (starting in SCH, followed by HAI and then ALB) for 5 days each in May 2014, December 2014, April 2015 and July 2015 (05.05.–23.05.14; 01.12.–12.12.14, 06.04.–24.04.15; 29.06.–17.07.15). Days of sampling were constrained by field permissions (weekends excluded) and logistics (9–12 sites per day).

As we did not discover any beetles in December and registered no removal at all, we excluded the December survey from further analysis and results.

In each site we monitored the dung beetle abundance and dung removal simultaneously for 48 h. To assess dung beetle abundance, we used dung-baited pitfall traps. To account for the beetles characteristic spectrum of dung resources available, we used six different dung baits consisting of three livestock and three game species, namely: cow (*Bos taurus* L., 1758), horse (*Equus caballus* L., 1758), sheep (*Ovis aries* L., 1758), red deer (*Cervus elaphus* L., 1758), wild boar (*Sus scrofa* L., 1758) and fox (*Vulpes vulpes* L., 1758). For removal rate experiments we used the same dung types (due to very low quantities of fox dung we were only able to use it for pitfall traps during the intensive survey). For both, pitfall traps and removal experiments, dung samples were collected from the same sources. Livestock dung was collected at the farm ‘Oberfeld’ in Darmstadt (cow and horse) and at a sheep farm in Darmstadt (sheep); all livestock animals were grazing in pastures for at least part of the day, cows were additionally provided hay. Game species dung has been collected in the wildlife park ‘Alte Fasanerie’ in Hanau (fox, wild boar and red deer) and at the zoo ‘Opel-Zoo’ in Kronberg (additional fox). Diets for the animals were as follows: cow: grazing on pasture, hay; horse and sheep: grazing on pasture; red deer: grass, hay, maize, fodder beet, lucerne pellets, apples, carrots; wild boar: pig food (Raiffeisen), bread, maize, fruit, vegetables, lucerne pellets, meat of cattle, fallow deer and red deer; fox: 60% meat (chicken, mice, rats, cattle), fruits, vegetables. Veterinarian medication (e.g. Ivermectin) can influence the treated animal’s dung and is known to have negative effects on dung beetle performance (Lumaret et al., 2012; Verdu et al., 2015). According to farmers and animal keepers, however, all animals involved in this study have not faced any veterinarian treatment for several weeks before dung collection. Therefore, we do not expect adverse reactions during removal activity.

After collecting samples in a sufficient amount, the dung was prepared in a lab either by filling dung in a tea bag and transferring the bait in a freezer bag or by filling freezer bags directly with dung for removal experiments. Afterwards the freezer bags where hermetically sealed, weighed and labelled. They were stored in a freezer at −20 °C until use, in order to prevent microbial decomposition, moulding or possible dung beetle activity (if small dung beetles had been accidentally collected in the dung).

2.2. Experimental design

Pitfall traps and removal rate experiments were placed on each site. Six pitfall traps (six dung types) were placed in a transect along the site margin, and in parallel five dung samples for removal assessment on the opposite side. Both, traps and dung samples were randomised on each site and separated by a distance of 10 m. For pitfall traps we used plastic cups (Ø 9.5 cm, height: 10 cm, vol.: 500 ml) and inserted dome lids with a hole (Ø 3 cm) as a funnel. The baits consisted of tea bags (Rubin, size S, Burgwedel, Germany) filled with dung (approx. 35 g for each bait), which were attached to a skewer by an elastic strap. When placing the traps, we took care that they were at ground level and had no barrier for walking beetles. The skewer was placed next to the pitfall, so that the inaccessible bait was approx. 10 cm above the center of the trap.

For removal experiments, we used dung with a fresh weight of approx. 220.7 (± 19.9) g of cow, 34.4 (± 3.8) g of horse, 50.5 (± 3.6) g of sheep, 32.6 (± 1.6) g of deer, 14.5 (± 1.4) g of fox and 47.6 (± 2.4) g of wild boar. All dung samples have been placed on cellulose paper. This method allowed us to verify whether missing dung can be assigned to dung beetle activity as we checked the paper for characteristic holes, which often occur by tunnelling species (*Geotrupes* and *Onthophagus*). We then calculated only the removal rate for those dung samples showing such holes or where the cellulose paper was destroyed. For other samples that may have missing amounts of dung (e.g. due to incomplete retrieval or activities of other animals), removal was set to zero. Furthermore we took only tunnelling species into account to analyse the correlation between beetle abundance in traps and removal per site. After 48 h pitfall traps were collected, trapped beetles were labelled (date, site-ID, dung type) and stored in a freezer at −20 °C. Dung samples of removal experiments were also collected, transferred into small paper bags, labelled (date, site-ID, dung type) and stored in a freezer at −20 °C. Caught dung beetles were identified to species level based on literature (Freude et al., 1969; Bunalski, 1999; Rössner, 2012) and with the help of taxonomic experts (see Acknowledgements). Removal samples were transferred into drying ovens and kept there at 60 °C for at least five days. Afterwards the dry weight for each dung sample was weighed (Mettler Toledo “EL 2001” (± 0.01 g), Columbus, Ohio) and noted for further calculations.

2.3. Data analysis

To examine the sampling completeness of occurring dung beetle species in each region and habitat (forests and grasslands), we calculated the estimated species richness test based on the Chao 1 index.

Temperature data (in °C) were measured with sensor stations installed within the Biodiversity Exploratories project on each site. Annual temperature time-series were used, to obtain the mean temperature over the 48 h dung/trap exposition time for each site at 10 cm above ground.

For land-use characterisation we used two habitat specific indices: land-use intensity (LUI) and forest management intensity (ForMI). LUI is based on grazing (*G*), i.e. the number of livestock units times the days of grazing per ha and year (see Fig. S5 for presence/absence of livestock and type of grazing), fertilisation (*F*), kg nitrogen applied per ha and year, and the frequency of mowing (*M*) per year. All parameters (*G*, *F*, *M*) are evaluated on annual basis (interview with farmers), which results in an updated land-use intensity index for each year, respectively. Furthermore, all factors were standardised per site *i* for a given year, relative to its mean within the corresponding model region *R* for that year. To reduce the impact of outliers and achieve a more even distribution, a square root-transformation was applied to the LUI, which is thus described as follows:

$$LUI = \sqrt{\frac{G_i}{G_R} + \frac{F_i}{F_R} + \frac{M_i}{M_R}}$$

Due to standardisation by ratios, the LUI is dimensionless. For more details see Blüthgen et al. (2012).

The ForMI is based on three parameters as well: the ratio of harvested tree volume to the sum of standing, harvested and dead wood volume (I_{harv}; a value of 0 describes no timber harvest in the last 30–40 years, a value of 1 a clear-cut site); the volume proportion of tree species that are not part of the natural forest composition, estimated as proportion of wood volume of non-native tree species to the sum of wood volume of all tree species (I_{nonat}; a value of 0 is a stand composed of natural forest vegetation only, 1 means that the whole stand consists of non-native tree-species); and the proportion of dead wood volume showing signs of saw cuts to the total amount of dead wood volume (I_{dwcut}; a value of 0 describes that all dead wood is a result of natural tree death, 1 that all dead wood is originated from management activity). These three parameters are summarised as:

$$ForMI = I_{harv} + I_{nonat} + I_{dwcut}$$

As the shift in tree species composition described in I_{nonat} comprises mainly coniferous species (*Picea abies* and *Pinus sylvestris*) that are not native in the study sites, we focus on the proportion of conifers below. Index data were assessed during forest inventory in 2008 and 2009 for living stands, in 2012 for I_{dwcut} (recording all dead wood > 25 cm on the 1 ha site), while I_{harv} was calculated based on allometric functions (Muukonen 2007). Since all components are calculated as proportions, they are dimensionless. For more details see (Kahl and Bauhus, 2014).

Due to the skewed distribution of the removal and biomass data we transformed each dataset to achieve normal distribution as described in Table 1. In addition we log-transformed the LUI index components (*G* and *F*) to avoid outliers in regressions.

Data were analysed with linear mixed-effects models (command ‘lme’) for the comprehensive survey and linear models (command ‘lm’) for intensive surveys, using the statistical software package R 2.15.3 (R Core Team, 2013) including the package´nlmé (Pinheiro et al., 2014). As response variable in the model, we either used the mean dung removal rate across all dung types, or the total dung beetle biomass across all dung types; hence dung types were pooled per site. Region and habitat-specific land-use indices or their components were employed as fixed factors in the model. The interaction terms served to assess whether the land use effects were consistent across regions or not. Dung beetle activities are known to vary with air temperature (Dortel et al., 2013) like other insect groups such as pollinators in the same study region (Kühsel and Blüthgen, 2015); to account for a potential temperature bias on dung beetle activities, we used temperature as a random factor. For the intensive surveys we used the factor ‘month’ rather than temperature, and site identity as random factor to account for replicated surveys. To summarise the responses of removal with increasing land-use intensity, we added slope values of linear regressions for each analysis. We estimated the per cent change along each gradient (full range from minimum to maximum land-use intensity) based on the slope and intercept from each model. When interaction terms were significant, we provided separate analyses for each region (Supplementary material: Tables S1 and S2).

Removal rates were defined by dry mass. By dividing dry weight by fresh weight we calculated the dry mass content (P_{DM}) for each dung type and sampling. This approach allowed us to estimate the dry mass removal for each dung sample placed in the field as:

$$removal\ \ (g) = (fw_{before} * P_{DM}) - dw_{after},$$

with fw_{before} being the fresh weight of a sample before exposition in the field and dw_{after} being the weight of the samples collected in the field after 48 h. To account for possible differences among collected dung types and collection dates, we determined water contents of 3–8 randomly selected samples for each dung type and each collection day and treated them as described above for removal samples, except that we did not expose them to the field. Average water contents (proportions) were for cow: 0.85 (± 0.01), sheep: 0.74 (± 0.01), horse: 0.69 (± 0.03), deer: 0.55 (± 0.02), fox: 0.41 (± 0.03), wild boar: 0.46 (± 0.02).

Table 1
Effects on overall dung removal rate (g dry mass per two days) and biomass (g dry weight) of dung beetles for (a) the comprehensive survey (n = 150 forests and 150 grassland sites) and (b) the intensive survey (n = 27 forests and 27 grasslands, four surveys each). Main effects (*F*-values) and significance (* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$) of linear mixed effect (LME) models are shown for habitat-specific indices (ForMI: Forest management intensity index, LUI: Land-use intensity index) and their components. The trend states the per cent increase or decrease of removal or biomass along the entire management gradient (from the minimum to the maximum gradient value), as predicted from the slope and intercept of the LME model.

			(a) comprehensive survey				(b) intensive survey			
			Removal [g/2d][a]		Biomass [g][b]		Removal [g/2d][a]		Biomass [g][b]	
Region	Habitat	Interaction	*F*	Trend	*F*	Trend	*F* [c]	Trend	*F* [c]	Trend
All	Forest	ForMI	0.3	+8.8%	1.0	+7.0%	0.0	+5.7%	0.2	+32.2%
		Region × ForMI	3.9*		2.4		3.6*		4.3*	
		TimberHarvest	5.3*	−19.6%	1.4	−35.6%	0.4	−0.7%	0.5	+28.5%
		Region × WoodHarvest	0.6		0.5		1.4		0.6	
		NonNativeTreeSpecies	6.5*	+22.3%	3.4	+22.3%	0.3	+13.6%	0.9	+67.5%
		Region × NonNativeTrees	6.3*		0.4		2.9		5.1**	
		DeadWoodWithSawCuts	0.2	+39.8%	0.5	+39.8%	0.0	+8.3%	0.1	+55.3%
		Region × DeadWood	0.8		2.6		1.7		1.9	
All	Grassland	LUI	0.3	−6.1%	2.5	−83.3	6.4**	−10.4%	1.8	+5.1%
		Region × LUI	0.6		1.0		0.9		2.0	
		log Grazing	7.7**	+5.7%	4.5*	+58.1%	2.4	+1.35%	0.4	−13.3%
		Region × (log Grazing)	0.2		0.4		0.4		0.4	
		log Fertilisation	2.3	−2.8%	5.6*	−20%	8.5**	−4.2%	2.0	+6.25%
		Region × (log Fertilisation)	0.1		1.0		0.2		0.9	
		Mowing	5.9*	−7.2%	8.9**	−41.6%	3.2	−7.9%	0.5	+26.6%
		Region × Mowing	0.2		0.3		0.9		1.5	

[a] Data square root transformed.
[b] Data log transformed.
[c] As the factor 'month' is used to take account for multiple surveys, the lowest *F*-value of all surveys is listed.

As the beetles' body size is correlated to their dung removal rate (Nervo et al., 2014), we translated the abundances of the beetles into biomass. We calculated the species-specific body mass, based on the mean dry weight in g (dried at 60 °C for 3 days) measured on a microbalance (Mettler Toledo "EL 2001" (± 0.01 g), Columbus, Ohio) for 3–10 randomly selected individuals per species across all regions. Species-specific dry mass was multiplied with the species' abundances per site to quantify the total biomass for further analysis.

3. Results

3.1. Dung beetle community

In the comprehensive survey, we sampled 8297 individuals from 26 species of dung beetles in 206 experimental sites, hence in 94 of the total 300 sites, no dung beetles were trapped. For the intensive survey we sampled 10483 individuals from 31 species in 54 experimental sites (each surveyed four times); the total number of species in all surveys was 33 (5 Geotrupidae, 10 *Onthophagus*, 18 *Aphodius* species). While the three most common species of the family Geotrupidae and 13 of the *Aphodius* species mainly occurred in forests and represent the vast majority of caught individuals, 8 species of the genus *Onthophagus* were predominantly found in grasslands (Fig. 1). Overall, dung beetle biomass was 10 times higher in forests than in grasslands in the Alb (Welch *t*-test, $t = 3.83$, $p < 0.001$), 20 times in the Hainich ($t = 5.62$, $p < 0.001$), up to 80 times in the Schorfheide ($t = 8.99$, $p < 0.001$) (Fig. 2b). Our surveys represented the dung beetle species pool very well: for the comprehensive survey the dung beetle richness in forests showed ≥ 87.5% sampling completeness when the recorded and estimated richness (Chao1) were compared in each region, and for grasslands the completeness was even 100%. For the intensive survey richness in forests showed ≥ 85.7% completeness, while the grasslands completeness was ≥ 86.9%.

In general, different management of the habitats showed contrasting effects on the beetles' biomass. While increasing by grazing intensity, the dung beetles' biomass decreased with the intensity of mowing and fertilisation in the comprehensive survey. For the intensive survey we found no significant effects, except at a regional level: only in the Alb, forests dominated by conifers had a higher dung beetles biomass (Figs. 3 and 4, Figs. S1 and S2, Tables S1 and S2).

3.2. Dung removal

Comprehensive survey: dung removal rates were 3.5 times higher in forests than in grasslands (Welch *t*-test, $t = 16.27$, $p < 0.001$) (Fig. 2a). Again, management showed contrasting effects: the proportion of non-native tree species (conifers) in forests and grazing intensity in grasslands as associated with an increase in dung removal (removal activity rises by +22.3% from forests with no non-native trees to monospecific conifer stands, and by +5.7% from no grazing to the maximum observed grazing intensity). The amount of timber harvest in forests and mowing events in grasslands, however, had significant negative effects, resulting in a decrease of removal activity by −19.6% and −7.2% along the range of these gradients, respectively (Figs. 5 and 6, Table 1).

Intensive survey: dung removal rates were 2.7 times higher in forests than in grasslands (Welch *t*-test, $t = 8.98$, $p < 0.001$). Regarding the land use effects, there was a decrease in removal due to fertilisation in grasslands (by −4.2%), whereas no significant effects were found in forests (Fig. 6, Figs. S3 and S4, Table 1).

Dung removal was highly correlated with biomass of caught beetles in pitfall traps of the same sites (Spearman rank test: $r_S = 0.78$, $p < 0.001$; all surveys combined). The increase in removal rate per additional beetle biomass was 0.52 g/g (slope of the log(removal + 1)/log(biomass + 1) relationship, $r^2 = 0.47$). For both surveys, removal on pastures did not differ among types of grazing (sheep, horse, cattle) (ANOVA, $F_{1,270} = 2.17$, $p = 0.14$).

4. Discussion

Dung beetles are important ecosystem service providers, but are also known to negatively respond to anthropogenic disturbances and land use across the globe (Nichols et al., 2007). Our study confirmed land-use effects on dung beetle activity and their ecosystem service in Central European cultural landscapes. In this region, many centuries of intense human activities have led to habitat conversion from forest to

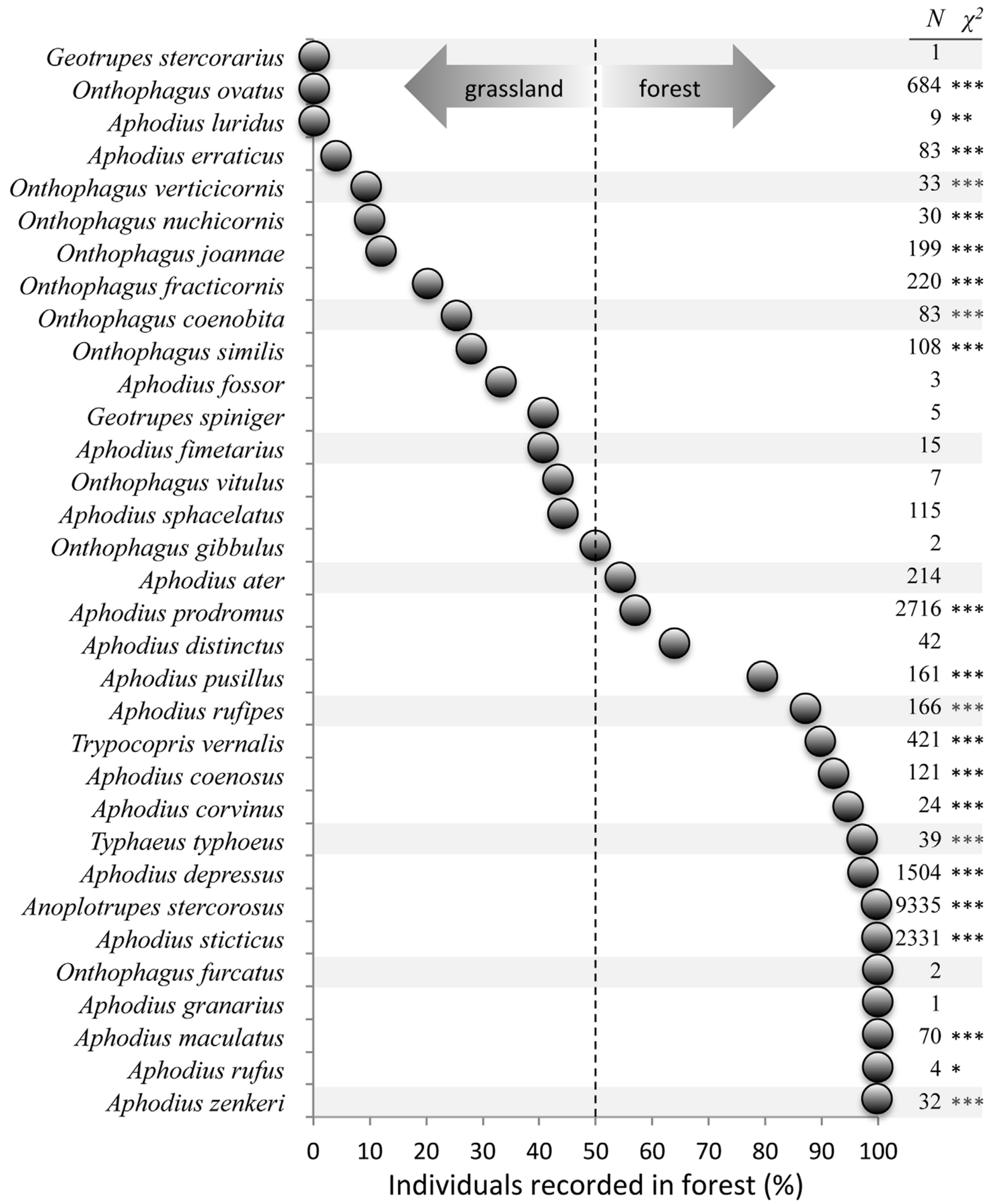

Fig. 1. Distribution of dung beetles between forest and grassland sites, given as the proportion of individuals of each species found in forests. *N* is the total number of individuals in both habitats, and the asterisks provide the significance of a habitat preference (χ^2-test against 50% for all species with $N \geq 5$: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

cultivated grassland and/or arable land. Forests – irrespective of their management – had several times higher beetle densities and dung removal rates than managed grasslands. While coniferous forests had significant positive effects on removal, timber harvest showed negative effects. In grasslands, beetles and their activity were positively affected by livestock densities, whereas mowing and fertilisation caused a decline of the beetles' biomass and removal activities. Whereas the difference between forests and grasslands was pervasive, the continuous land-use intensity gradients within forests or grasslands showed much weaker trends. Significant gradual responses were detected only in the comprehensive survey, but not confirmed in the intensive survey on a subset of the sites except the negative effect of fertilisation on removal. Therefore, a high number of spatially independent replicates is important for detecting such within-habitat responses.

Our assessment of dung removal within two days predominantly reflected the activities of the tunnellers, i.e. the family of Geotrupidae and the genus *Onthophagus*. Larger time scales of several weeks, up to months, give additional taxa the opportunity to contribute in removal activity, which highlights the complexity of multiple services for ecosystem functioning (Nervo et al., 2017). Still, we focused on the two tunnelling taxa as they primarily bury dung (Anduaga, 2004; Walters, 2008) and are responsible for most of the removal within a short time frame of two days (Nervo et al., 2014). Additionally, the tunnellers represented 60% of the dung beetle individuals collected and 96% of the overall beetle biomass.

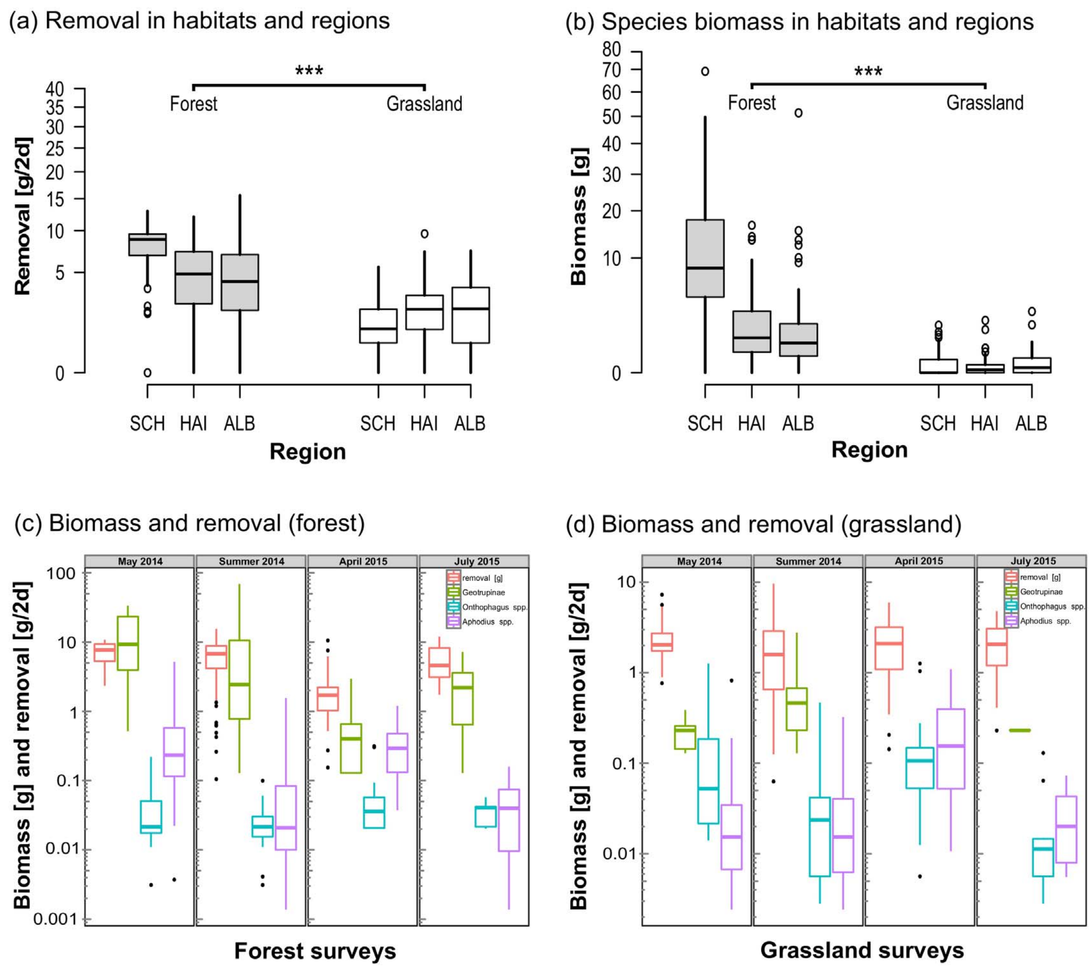

Fig. 2. Overview for all surveys. Mean dung removal (g dry weight per two days) (a) and mean biomass of caught beetles (b) for each region and habitat and comparison of mean dung removal with mean biomass of caught beetles for each survey (c) and (d). Note that the y-axis of (a) is square root transformed, while the y-axis of (b), (c) and (d) is log-transformed.

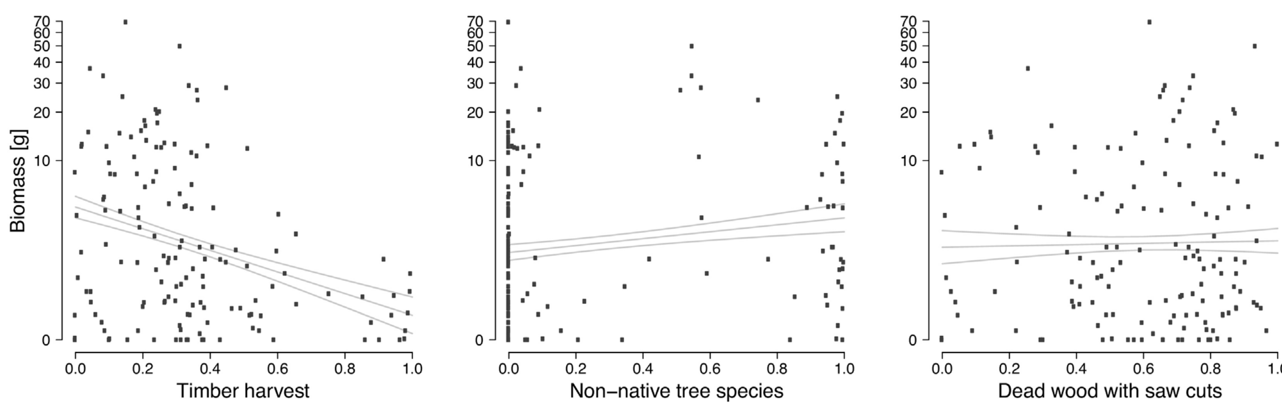

Fig. 3. Biomass of individuals (g dry mass) collected during the comprehensive survey (n = 150 sites) plotted against components of forest management intensity index (ForMI): proportion of timber harvest, proportion of non-native tree species and proportion of dead wood with saw cuts. Note that the y-axis is log-transformed.

4.1. Habitat preference

Across the regions we found clear indications of habitat differentiation. The tunnelling species *Anoplotrupes stercorosus* was almost exclusively caught in forests where it was the most dominant species. Moreover, *Trypocopris vernalis* mostly occurred in coniferous forests. Conversely, *Onthophagus fracticornis, O. joannae* and *O. ovatus* constituted the majority of Scarabaeinae in grasslands. Peyras et al. (2013) highlighted distinct species-specific responses towards habitat fragmentation for dung beetles. In addition – depending on species, resource distribution and environmental conditions – habitat preferences can lead to distinct spatial mosaics of dung beetle communities and edge effects at habitat margins (Ries and Sisk, 2004). As dung beetles typically utilise fresh dung piles which emit higher amounts of volatiles and provide suitable consistence for handling, the exposition of small dung piles on unshaded grasslands can cause higher dehydration and thus result in lower attractivity and removal. Since beetle abundance, biomass and removal rates were strongly correlated (consistent with the

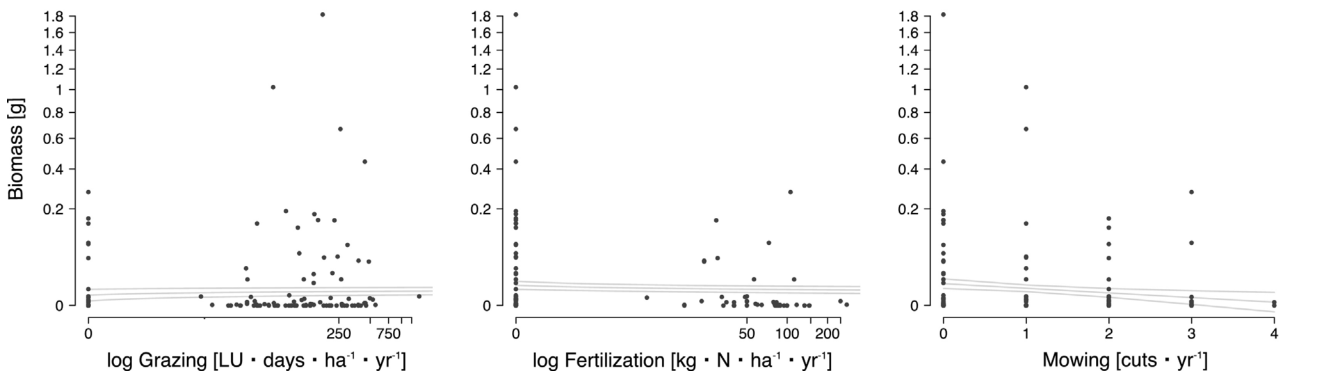

Fig. 4. Biomass of individuals (g dry mass) collected during the comprehensive survey (n = 150 sites) plotted against components of land-use intensity index (LUI): grazing, fertilisation and mowing. Note that y-axis, grazing and fertilisation (x-axis) is log-transformed.

findings by Tixier et al., 2015), the lower beetle densities in grasslands corresponded to a much lower removal compared to forests. In consequence, tree removal and open canopies can be crucial for a substantially reduced dung removal service.

4.2. Land use and forest management

Depending on different landowners, regulations and goals, forest management and thus the level of disturbance differs substantially. Unlike the regular annual management activities in grassland, however, harvesting in forests is typically followed by long periods of regrowth; it thus seems likely that mobile dung beetles follow the forest recovery and succession. In fact, the spatial distribution of the dung-producing mammals determines the dung beetles occurrence (Kuhn, 2010), and the beetles are able to sense dung volatiles at a great distance (Hanski and Cambefort, 1991). Rembialkowska et al. (1982) described the use of foliage and coniferous litter for brood balls in *Geotrupes stercorosus*, highlighting the ability of dung beetles to switch to other resources when dung is scant. By this, and because of the dung beetles shift from saprophagy to coprophagy (Cambefort, 1991), forests provide additional resources besides dung, sustaining the survival of locally high numbers of beetles. Large areas of fir and pine forests, often planted after the second world war for quick reforestation and timber (Elling, 2007), can sustain particularly high densities of *Anoplotrupes* e.g. in the Schorfheide. Such coniferous forests may be particularly attractive for game species by providing a more sheltered habitat, but also because of its continuous management and a strong development of understory vegetation (e.g. ferns, raspberries, wild strawberry, blueberries) that serves as food for deer. As there is a strict management regime for game species in Germany, furthermore, there is strong anthropogenic pressure on the distribution of wildlife, either by physical barriers, such as fences, or by hunting (Arnold et al., 2015). In consequence, attractiveness of certain habitats and regions for dung beetles may respond to local game species populations. For the comprehensive survey, we showed management effects on dung removal rates in forests, yet no effects on dung beetle biomass. Because of the sheer abundance and widespread distribution of *Anoplotrupes stercorosus* (representing 94% of all tunnelling specimens in forests), the functional contribution of this single species may largely compensate for losses of some less abundant species.

In pastures, the density of livestock units per hectare has a positive effect on dung beetle activities, most likely driven by a higher availability of dung. Although intensive grazing can reduce dung beetle abundance in Ireland (Hutton and Giller, 2003), the net effect was positive in our study (Table 1, Fig. S5 for livestock presence/absence on sites). On the other hand, heavy disturbances such as mowing might disrupt the establishment of dung beetle communities. Furthermore, pastures without permanent grazing (e.g. hayfields) offer poor conditions due to insufficient resources and by this are incapable to balance adverse conditions like higher temperatures (drought stress) and increased predation risk. Since different dung beetle species do have seasonal niches (Figs. 2c and d) (Waßmer, 1994), depending on the management, pastures may lack of attractiveness during different seasons and by this might cause the absence of certain taxa. Fertilisation and mowing are usually coupled, highly correlated and thus difficult to disentangle in their effects on dung beetles. Enhanced fertilisation results in higher productivity and increasing mowing frequency, yet another trade-off to species with decreased tolerance towards stress.

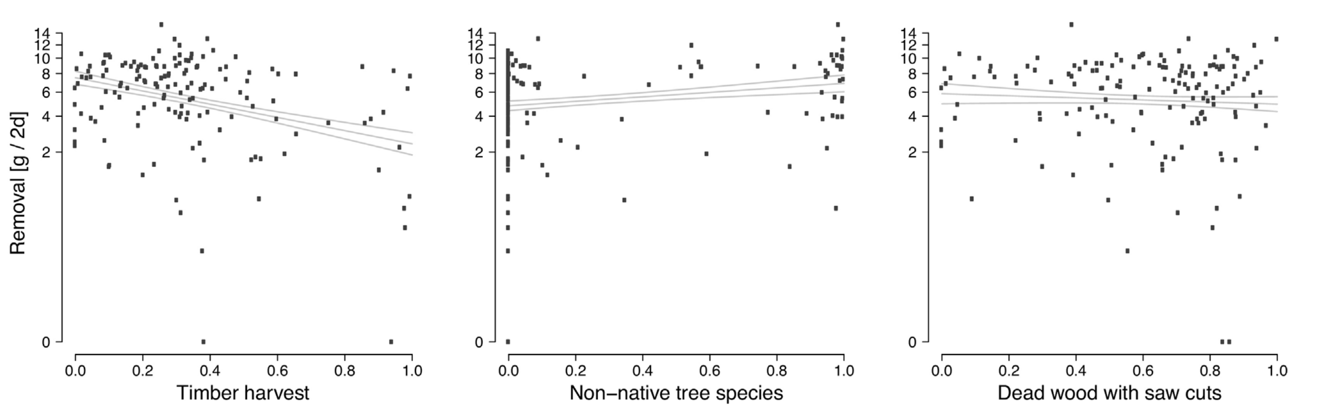

Fig. 5. Mean dung removal (g dry weight per two days) on forest sites (comprehensive survey, n = 150) plotted against the components of forest management intensity index (ForMI): proportion of timber harvest, proportion of non-native tree species and proportion of dead wood with saw cuts. Note that the y-axis is square root transformed.

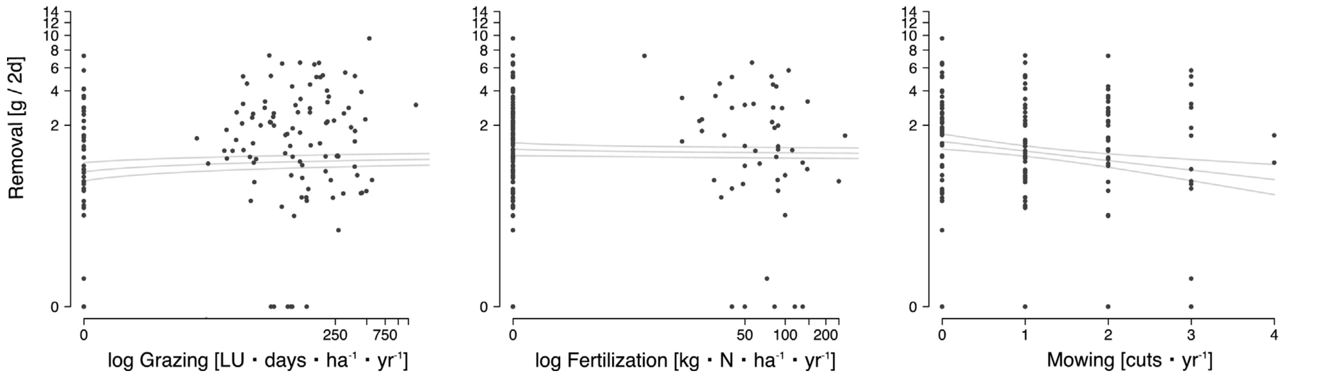

Fig. 6. Mean dung removal (g dry weight per two days) on grassland sites (comprehensive survey, n = 150) plotted against the components of land-use intensity index (LUI): grazing, fertilisation and mowing. Note that the y-axis is square root transformed, while grazing and fertilisation (x-axis) is log-transformed.

5. Conclusion

We were able to show on a species level habitat preference for dung beetles in a large-scale framework in three regions across Germany. Habitat differentiation has been described in the past, but there is a lack of information why certain dung beetle taxa differ in such great numbers between forests and grasslands (e.g. for the Schorfheide, 80 times higher beetle densities were found in forests than in grasslands). The strong habitat preferences among different genera suggest that there are other important environmental drivers besides anthropogenic influence, such as climatic conditions or soil quality, which seems incentive for further investigations.

For the comprehensive survey we were able to show broad effects of grassland and forest management intensity, affecting dung beetles and their ecosystem service negatively, likewise in forests and grassland. Since our analysis showed contrasting local effects − proportion of non-native tree species versus timber harvest, grazing versus fertilisation and mowing − dung beetle activities were relatively balanced along the aggregate land-use intensity gradients (ForMI, Kahl and Bauhus, 2014 and LUI, Blüthgen et al., 2012) (Figs. S6 and S7), and impacts are complex. Dung beetles seem to be able to compensate certain amounts of gradual disturbance by abundance, mobility or seasonal occurrence. However, as our study was conducted in regions that include protected areas and relatively heterogeneous landscapes, not dominated by intensive arable land, many of the broad risks for dung beetle communities are not covered. Major disturbances and regime shifts, particularly deforestation or conversion to arable land, pesticide application or lactone treatments of livestock are likely to cause a much stronger decrease of dung beetle densities, resulting in declining removal activity (Nichols et al., 2007; Newbold et al., 2015).

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.agee.2017.04.010.

References

Allan, E., Manning, P., Alt, F., Binkenstein, J., Blaser, S., Bluethgen, N., Bohm, S., Grassein, F., Holzel, N., Klaus, V.H., Kleinebecker, T., Morris, E.K., Oelmann, Y., Prati, D., Renner, S.C., Rillig, M.C., Schaefer, M., Schloter, M., Schmitt, B., Schoning, I., Schrumpf, M., Solly, E., Sorkau, E., Steckel, J., Steffen-Dewenter, I., Stempfhuber, B., Tschapka, M., Weiner, C.N., Weisser, W.W., Werner, M., Westphal, C., Wilcke, W., Fischer, M., 2015. Land use intensification alters ecosystem multifunctionality via loss of biodiversity and changes to functional composition. Ecol. Lett. 18, 834–843.

Anderson, J., Merritt, R., Loomis, E., 1984. The insect-free cattle dropping and its relationship to increased dung fouling of rangeland pastures. J. Econ. Entomol. 77, 133–141.

Anduaga, S., 2004. Impact of the activity of dung beetles (Coleoptera: scarabaeidae: Scarabaeinae) inhabiting pasture land in Durango, Mexico. Environ. Entomol. 33, 1306–1312.

Arnold, J.M., Greiser, G., Kampmann, S., Martin, I., 2015. Status Und Entwicklung Ausgewählter Wildtierarten in Deutschland. Jahresbericht 2014. Wildtier-Informationssystem der Länder Deutschlands (WILD), Deutscher Jagdverband, Berlin.

Bang, H.S., Lee, J.H., Kwon, O.S., Na, Y.E., Jang, Y.S., Kim, W.H., 2005. Effects of paracoprid dung beetles (Coleoptera: scarabaeidae) on the growth of pasture herbage and on the underlying soil. Appl. Soil Ecol. 29, 165–171.

Barragan, F., Moreno, C.E., Escobar, F., Halffter, G., Navarrete, D., 2011. Negative impacts of human land use on dung beetle functional diversity. PLoS One 6, e17976.

Blüthgen, N., Dormann, C.F., Prati, D., Klaus, V.H., Kleinebecker, T., Holzel, N., Alt, F., Boch, S., Gockel, S., Hemp, A., Muller, J., Nieschulze, J., Renner, S.C., Schoning, I., Schumacher, U., Socher, S.A., Wells, K., Birkhofer, K., Buscot, F., Oelmann, Y., Rothenwohrer, C., Scherber, C., Tscharntke, T., Weiner, C.N., Fischer, M., Kalko, E.K.V., Linsenmair, K.E., Schulze, E.D., Weisser, W.W., 2012. A quantitative index of land-use intensity in grasslands: integrating mowing, grazing and fertilization. Basic Appl. Ecol. 13, 207–220.

Bornemissza, G., 1960. Could dung eating insects improve our pastures. J. Aust. Inst. Agric. Sci. 26, 54–56.

Bunalski, M., 1999. Die Blatthornkäfer Mitteleuropas (Coleoptera, Scarabaeoidea): Bestimmung, Verbreitung, Ökologie. František Slamka, Slovakia.

Cambefort, Y., 1991. From saprophagy to coprophagy. In: Hanski, I., Cambefort, Y. (Eds.), Dung Beetle Ecology. Princeton University Press, Princeton, NJ, USA, pp. 22–35.

Chin, K., Gill, B.D., 1996. Dinosaurs, dung beetles, and conifers: participants in a Cretaceous food web. Palaios 280–285.

Davis, A.L.V., Scholtz, C.H., Philips, T.K., 2002. Historical biogeography of scarabaeine dung beetles. J. Biogeogr. 29, 1217–1256.

Dormont, L., Rapior, S., McKey, D.B., Lumaret, J.P., 2007. Influence of dung volatiles on the process of resource selection by coprophagous beetles. Chemoecology 17, 23–30.

Dortel, E., Thuiller, W., Lobo, J.M., Bohbot, H., Lumaret, J.P., Jay-Robert, P., 2013. Potential effects of climate change on the distribution of Scarabaeidae dung beetles in Western Europe. J. Insect Conserv. 17, 1059–1070.

Elling, W., 2007. Schädigung Von Waldökosystemen: Auswirkungen Anthropogener Umweltveränderungen Und Schutzmaßnahmen. Elsevier, Spektrum Akad. Verlag, Germany.
Estrada, A., Coates-Estrada, R., 2002. Dung beetles in continuous forest, forest fragments and in an agricultural mosaic habitat island at Los Tuxtlas, Mexico. Biodivers. Conserv. 11, 1903–1918.
Feer, F., Hingrat, Y., 2005. Effects of forest fragmentation on a dung beetle community in French Guiana. Conserv. Biol. 19, 1103–1112.
Fincher, G.T., 1973. Nidification and reproduction of Phanaeus spp. in 3 textural classes of soil (Coleoptera: scarabaeidae). Coleopts Bull. 27, 33–37.
Fischer, M., Bossdorf, O., Gockel, S., Hänsel, F., Hemp, A., Hessenmöller, D., Korte, G., Nieschulze, J., Pfeiffer, S., Prati, D., 2010. Implementing large-scale and long-term functional biodiversity research: the biodiversity exploratories. Basic Appl. Ecol. 11, 473–485.
Freude, H., Harde, K.W., Lohse, G.A., Klausnitzer, B., 1969. Die Käfer Mitteleuropas. 8. Teredilia, Heteromera Lamellicornia. Spektrum Akad. Verlag.
Halffter, G., Matthews, E.G., 1966. The natural history of dung beetles of the subfamily Scarabaeinae (Coleoptera: scarabaeidae). A monograph. Folia Entomol. Mex. 12/14, 1–312.
Hanski, I., Cambefort, Y., 1991. Dung Beetle Ecology. Princeton University Press, Princeton, NJ, USA.
Hanski, I., Koivulehto, H., Cameron, A., Rahagalala, P., 2007. Deforestation and apparent extinctions of endemic forest beetles in Madagascar. Biol. Lett. 3, 344–347.
Herzog, F., Steiner, B., Bailey, D., Baudry, J., Billeter, R., Bukacek, R., De Blust, G., De Cock, R., Dirksen, J., Dormann, C.F., De Filippi, R., Frossard, E., Liira, J., Schmidt, T., Stockli, R., Thenail, C., van Wingerden, W., Bugter, R., 2006. Assessing the intensity of temperate European agriculture at the landscape scale. Eur. J. Agron. 24, 165–181.
Hutton, S.A., Giller, P.S., 2003. The effects of the intensification of agriculture on northern temperate dung beetle communities. J. Appl. Ecol. 40, 994–1007.
Kühsel, S., Blüthgen, N., 2015. High diversity stabilizes the thermal resilience of pollinator communities in intensively managed grasslands. Nat. Commun. 6.
Kahl, T., Bauhus, J., 2014. An index of forest management intensity based on assessment of harvested tree volume, tree species composition and dead wood origin. Nat. Conserv. Bulgaria 15–27.
Kuhn, K., 2010. Kartierung der dungbewohnenden Käferarten im beweidungsgebiet des NSG stadtwald augsburg. Berichte des Naturwissenschaftlichen Vereins für Schwaben 114, 102–115.
Lumaret, J.P., Errouissi, F., Floate, K., Rombke, J., Wardhaugh, K., 2012. A review on the toxicity and non-target effects of macrocyclic lactones in terrestrial and aquatic environments. Curr. Pharm. Biotechnol. 13, 1004–1060.
Manning, P., Slade, E.M., Beynon, S.A., Lewis, O.T., 2016. Functionally rich dung beetle assemblages are required to provide multiple ecosystem services. Agric. Ecosyst. Environ. 218, 87–94.
Martín-Piera, F., Lobo, J., 1995. Diversity and ecological role of dung beetles in Iberian grassland biomes. In: McCracken, D.I., Bignal, E.M., Wenlock, S.E. (Eds.), Farming on the Edge: the Nature of Traditional Farmland in Europe. Joint Nature Conservation Committee, Peterborough, pp. 147–153.
Nervo, B., Tocco, C., Caprio, E., Palestrini, C., Rolando, A., 2014. The effects of body mass on dung removal efficiency in dung beetles. PLoS One 9, e107699.
Nervo, B., Caprio, E., Celi, L., Lonati, M., Lombardi, G., Falsone, G., Iussig, G., Palestrini, C., Said-Pullicino, D., Rolando, A., 2017. Ecological functions provided by dung beetles are interlinked across space and time: evidence from N-15 isotope tracing. Ecology 98, 433–446.
Newbold, T., Hudson, L.N., Hill, S.L.L., Contu, S., Lysenko, I., Senior, R.A., Borger, L., Bennett, D.J., Choimes, A., Collen, B., Day, J., De Palma, A., Diaz, S., Echeverria-Londono, S., Edgar, M.J., Feldman, A., Garon, M., Harrison, M.L.K., Alhusseini, T., Ingram, D.J., Itescu, Y., Kattge, J., Kemp, V., Kirkpatrick, L., Kleyer, M., Correia, D.L.P., Martin, C.D., Meiri, S., Novosolov, M., Pan, Y., Phillips, H.R.P., Purves, D.W., Robinson, A., Simpson, J., Tuck, S.L., Weiher, E., White, H.J., Ewers, R.M., Mace, G.M., Scharlemann, J.P.W., Purvis, A., 2015. Global effects of land use on local terrestrial biodiversity. Nature 520, 45–50.
Nichols, E., Larsen, T., Spector, S., Davis, A.L., Escobar, F., Favila, M., Vuline, K., 2007. Global dung beetle response to tropical forest modification and fragmentation: a quantitative literature review and meta-analysis. Biol. Conserv. 137, 1–19.
Nichols, E., Spector, S., Louzada, J., Larsen, T., Amequita, S., Favila, M.E., Network, S.R., 2008. Ecological functions and ecosystem services provided by Scarabaeinae dung beetles. Biol. Conserv. 141, 1461–1474.
Nikolajev, G.V., Dong, R., 2010. New genus of the subfamily geotrupinae (Coleoptera: scarabaeoidea: Geotrupindae) from the Jehol Biota. Acta Geol. Sin. Engl. 84, 673–675.
Peyras, M., Vespa, N.I., Bellocq, M.I., Zurita, G.A., 2013. Quantifying edge effects: the role of habitat contrast and species specialization. J. Insect Conserv. 17, 807–820.
Philips, T.K., 2011. The evolutionary history and diversification of dung beetles. Ecology and evolution of dung beetles. In: Simmons, L.W., Ridsdill-Smith, T.J. (Eds.), Ecology and Evolution of Dung Beetles. Wiley-Blackwell, Chichester, pp. 21–46.
Pinheiro, J., Bates, D., DebRoy, S., Sarkar, D., 2014. R Core Team 2014. nlme: linear and nonlinear mixed effects models. R package version 3. 1–117.
Rössner, E., 2012. Die Hirschkäfer Und Blatthornkäfer Ostdeutschlands (Coleoptera: Scarabaeoidea). Verein der Freunde und Förderer des Naturkundemuseums Erfurt, Germany.
Reidsma, P., Tekelenburg, T., van den Berg, M., Alkemade, R., 2006. Impacts of land-use change on biodiversity: an assessment of agricultural biodiversity in the European Union. Agric. Ecosyst. Environ. 114, 86–102.
Ridsdill-Smith, T.J., Edwards, P.B., 2011. Biological control: ecosystem functions provided by dung beetles. In: Simmons, L.W., Ridsdill-Smith, T.J. (Eds.), Ecology and Evolution of Dung Beetles. Wiley-Blackwell, Chichester, pp. 245–266.
Ries, L., Sisk, T.D., 2004. A predictive model of edge effects. Ecology 85, 2917–2926.
Romero-Alcaraz, E., Avila, J.M., 2000. Effect of elevation and type of habitat on the abundance and diversity of Scarabaeoid dung beetle (Scarabaeoidea) assemblages in a Mediterranean area from Southern Iberian Peninsula. Zool. Stud. 39, 351–359.
Roslin, T., Viljanen, H., 2011. Dung beetle populations: structure and consequences. In: Simmons, L.W., Ridsdill-Smith, T.J. (Eds.), Ecology and Evolution of Dung Beetles. Wiley-Blackwell, Chichester, pp. 220–244.
Schmitt, T., Krell, F.T., Linsenmair, K.E., 2004. Quinone mixture as attractant for necrophagous dung beetles specialized on dead millipedes. J. Chem. Ecol. 30, 731–740.
Scholtz, C.H., Davis, L.V., Kryger, U., 2009. Evolutionary Biology and Conservation Ecology of Dung Beetles. Pensoft Publishers, Sofia, Bulgaria.
Soliveres, S., van der Plas, F., Manning, P., Prati, D., Gossner, M.M., Renner, S.C., Alt, F., Arndt, H., Baumgartner, V., Binkenstein, J., Birkhofer, K., Blaser, S., Blüthgen, N., Boch, S., Böhm, S., Börschig, C., Buscot, F., Diekötter, T., Heinze, J., Hölzel, N., Jung, K., Klaus, V.H., Kleinebecker, T., Klemmer, S., Krauss, J., Lange, M., Morris, E.K., Müller, J., Oelmann, Y., Overmann, J., Pašalić, E., Rillig, M.C., Schaefer, H.M., Schloter, M., Schmitt, B., Schöning, I., Schrumpf, M., Sikorski, J., Socher, S.A., Solly, E.F., Sonnemann, I., Sorkau, E., Steckel, J., Steffan-Dewenter, I., Stempfhuber, B., Tschapka, M., Türke, M., Venter, P.C., Weiner, C.N., Weisser, W.W., Werner, M., Westphal, C., Wilcke, W., Wolters, V., Wubet, T., Wurst, S., Fischer, M., Allan, E., 2016. Biodiversity at multiple trophic levels is needed for ecosystem multifunctionality. Nature 536, 456–459.
Spector, S., 2006. Scarabaeine dung beetles (coleoptera: scarabaeidae: Scarabaeinae): An invertebrate focal taxon for biodiversity research and conservation. Coleopts Bull. 60, 71–83.
Tixier, T., Bloor, J.M.G., Lumaret, J.P., 2015. Species-specific effects of dung beetle abundance on dung removal and leaf litter decomposition. Acta Oecol. 69, 31–34.
Tocco, C., Negro, M., Rolando, A., Palestrini, C., 2013. Does natural reforestation represent a potential threat to dung beetle diversity in the Alps. J. Insect Conserv. 17, 207–217.
Vander Wall, S.B., 1990. Food Hoarding in Animals. University of Chicago Press.
Verdu, J.R., Cortez, V., Ortiz, A.J., Gonzalez-Rodriguez, E., Martinez-Pinna, J., Lumaret, J.P., Lobo, J.M., Numa, C., Sanchez-Pinero, F., 2015. Low doses of ivermectin cause sensory and locomotor disorders in dung beetles. Sci. Rep. 5, 13912.
Waßmer, T., 1994. Seasonality of coprophagous beetles in the Kaiserstuhl area near Freiburg (SW-Germany) including the winter months. Acta Oecol. 15, 607–631.
Walters, C., 2008. The Greatest Invention: Dung Beetles and a Cowman's Profits. Acres, Austin, Texas, USA.
Whipple, S.D., Hoback, W.W., 2012. A comparison of dung beetle (Coleoptera: scarabaeidae) attraction to native and exotic mammal dung. Environ. Entomol. 41, 238–244.
Wu, X.W., Duffy, J.E., Reich, P.B., Sun, S.C., 2011. A brown-world cascade in the dung decomposer food web of an alpine meadow: effects of predator interactions and warming. Ecol. Monogr. 81, 313–328.
Zamora, J., Verdú, J.R., Galante, E., 2007. Species richness in mediterranean agroecosystems: spatial and temporal analysis for biodiversity conservation. Biol. Conserv. 134, 113–121.